MASTER ENGINEERING DESIGN DOSSIER

140.40 kWp Grid-Connected Rooftop Solar PV Installation

Al Noor Packaging Factory — Dubai, UAE

Prepared For: Al Noor Packaging LLC
Location: Al Quoz Industrial Area, Dubai, United Arab Emirates
Grid Connection Base: DEWA 3-Phase, 400 V, 50 Hz Utility Infrastructure
Engineering Framework: Shams Dubai DRRG Regulations Version 2.0 Compliance
Document Control Code: AN-PV-PRJ-2026-REV3
Date of Issue: August 2026

1. Phase 1 — Spatial & Structural Roof Suitability

A rigorous spatial analysis was performed on the reinforced-concrete roof deck of the primary manufacturing building to establish structural margins and physical boundary limits before design configuration.

Formula 1.1 — Gross Plan Window Envelope:

*A_gross = Length * Width = 62.0 m * 38.0 m = 2,356.0 m²*

Formula 1.2 — Net Usable Installation Area:

To ensure full safety access compliance, a mandatory 1.5-meter outer boundary setback corridor was subtracted from all perimeter faces. Additionally, unalterable mechanical water-tank housings (10m x 6m Eastern zone) and HVAC infrastructure groups (8m x 5m North-West zone) were completely removed from layout calculations:
A_net = A_gross - Setbacks - Mechanical Obstructions = 1,965.0 m²

Formula 1.3 — Structural Equipment Load Threshold:

The factory roof profile features a maximum pre-certified civil dead load capacity limit of 25.0 kg/m² for additional permanent mechanical equipment. The cumulative weight distribution of our optimized plant layout tracks safely beneath this ceiling:
Distributed System Mass = Modules (22.0 kg) + Racking + Ballast Block Matrix = 11.45 kg/m² < 25.0 kg/m² (SAFE)

Formula 1.4 — Dynamic Wind Design Pressure (q_z):

Per ASCE 7-22 codes tracking Dubai localized meteorological criteria, the extreme wind velocity envelope is calculated based on a 3-second gust design profile of 47 m/s, yielding:
*q_z = 0.613 * V² * K_z * K_zt * K_d = 1,197 N/m²*

2. Phase 2 — Electrical Sizing & Component Configuration

The facility's historical billing records present a gross cumulative energy import requirement of 1,098,000 kWh per annum. This represents a mean baseline demand profile of 3,008.2 kWh per day.

Formula 2.1 — Active Industrial Daytime Demand Baseline:

The manufacturing floor runs a robust operational shift from Monday to Saturday, between 08:00 and 18:00, compiling exactly 3,120 high-load daytime production hours annually. This maps out a steady instantaneous load floor:
P_daytime = Annual Generation Target / Operating Hours = 1,098,000 kWh / 3,120 h = 351.92 kW

Formula 2.2 — Idealized Offsetting Target vs. Infrastructure Constraints:

While a symmetrical 100% net bill offset requires a massive theoretical array capacity of 683.68 kWp under a localized regional 5.5 Peak Sun Hours (PSH) environment, the plant capacity is limited by facility connection

walls. The physical grid connection incomer contains a 630A main circuit breaker ceiling that imposes a maximum continuous solar power injection wall of 350 kW AC.

By deploying a single high-efficiency Huawei SUN2000-110KTL-M2 inverter unit (110 kW nominal capacity), the plant output profile aligns perfectly beneath the grid threshold. Furthermore, because the plant's maximum hourly generation (~110 kW) stays significantly below the active daytime base load (351.92 kW), 100% of the generated green energy is instantly consumed on-site. This satisfies the utility's mandatory zero-export / export limitation requirement and completely eliminates the risk of wasteful power dumping via active curtailment devices.

3. Phase 3 — Thermal String Capacity Boundaries

To shield the inverter's power electronic channels from overvoltage destruction while ensuring sufficient voltage to stay online during ambient climate extremes, thermal string length calculations were hand-calculated using the parameters of the selected LONGi 585W hardware asset.

Formula 3.1 — Maximum Winter Series String Length Safety Ceiling (N_max):

During worst-case cold winter mornings, regional ambient air drops to 5°C. As temperature decreases, the solar panel open-circuit voltage spikes inversely via its temperature coefficient (γ_{voc} = -0.23%/°C):
$V_{oc(Coldest)} = V_{oc(STC)} * [1 + (T_{min} - 25°C) * \gamma_{voc}] = 52.36V * [1 + (5 - 25) * (-0.0023)] = 54.36\ V$
Applying the rigid mathematical Floor function against the Huawei inverter's maximum allowed DC input envelope (1,100 V DC) locks in the string safety limit:
$N_{max} = Floor(V_{max_inverter} / V_{oc(Coldest)}) = Floor(1100\ V / 54.36\ V) = Floor(20.23) = 20\ Panels\ Max$

Formula 3.2 — Minimum Summer Series String Length Floor (N_min):

During peak summer heat waves, external air temperatures reach 50°C. Dark silicon absorption surfaces experience extreme thermal gains, driving internal cell heat up to 75°C (50°C ambient + 25°C radiant offset). This heat degrades module operating voltage:
$V_{mpp(Hottest)} = V_{mpp(STC)} * [1 + (T_{cell} - 25°C) * \gamma_{voc}] = 44.21V * [1 + (75 - 25) * (-0.0023)] = 39.12\ V$
To keep the strings generating full rated power, the combined rows must clear the Huawei inverter's minimum full-load MPPT tracking voltage requirement (540 V) even during peak summer heat waves:
$N_{min} = Ceil(V_{mpp_min_full_load} / V_{mpp(Hottest)}) = Ceil(540\ V / 39.12\ V) = Ceil(13.80) = 14\ Panels\ Min$

4. Phase 4 — Inter-Row Spacing & Aerodynamic Wind Analysis

To optimize rooftop space utilization and minimize wind sail vulnerabilities, the system is designed in a landscape (horizontal) orientation on single-row L1 click-fit frames. Turning the panels sideways shifts the physical dimension running up the incline slope to the module's width parameter (1.134 m) rather than its length (2.278 m).

Formula 4.1 — Inter-Row Shading Pitch Boundaries:

To guarantee zero inter-row shading at solar noon during the winter solstice (December 21st, solar altitude angle α = 41.5° where tan(41.5°) = 0.8847), row spacing is governed by trigonometric parameters:

*D = L_mod * cos(β) + [(L_mod * sin(β)) / tan(α)]*

A side-by-side comparison evaluated the structural performance of the 10° and 15° landscape tilt choices:

Engineering Layout Metric	Option A: 10° Landscape Tilt	Option B: 15° Landscape Tilt
Horizontal Footprint Projection (P_horiz)	1.117 meters	1.095 meters
Minimum Clear Shading Walkway (D_clear)	22.3 cm (Too Tight for Cleaning Access)	33.2 cm (Optimal Clear Shading Pass)
Total Row-to-Row Pitch Distance (D)	**1.340 meters**	**1.427 meters**
Module Structural Apex Height (h)	19.7 cm (Ultra-Low Profile)	29.4 cm (Ultra-Low Profile)
Aerodynamic Net Pressure Coefficient (C_n)	-0.60 (Low Suction Lift Effect)	-0.85 (Medium Sail Suction Effect)
Wind Uplift Suction Force Per Module	**1,576.8 Newtons (~161 kg)**	**2,233.8 Newtons (~228 kg)**
Total Plant Cumulative Upward Lift	**38.58 Metric Tons**	**54.65 Metric Tons**

Engineering Access Adjustment Note: While a 10° or 15° tilt satisfies minimum shadow spacing at less than 1.5 meters row pitch, implementing the raw mathematical clear path (22cm to 33cm) makes physical cleaning walkways impossible. Therefore, the racking layout design manually overrides this boundary to a fixed 0.50m or 0.60m walkway gap, providing safe clearance for maintenance crews while remaining entirely free of winter shadows.

5. Phase 5 — Symmetrical System Balance Architecture

By choosing our absolute winter maximum capacity limit of 20 panels per string exactly, we establish a perfectly symmetrical, cost-optimized electrical schematic. Dividing our total target of 240 panels creates exactly 12 identical series strings.

Formula 5.1 — DC-to-AC Inverter Loading Ratio:

*The gross capacity of our array fields represents 240 modules * 585 W = 140.40 kWp DC. Wiring this straight into our single 110 kW AC string inverter gives us a perfect sizing configuration:*
Inverter Loading Ratio = 140.40 kWp DC / 110.00 kW AC = 1.2766 (Optimal Performance Window)

This 1.28 loading ratio acts as a strategic performance layer. It over-dimensions the DC field to compensate for real-world environmental losses (such as Dubai's high ambient heat and desert dust soiling), flattening the generation curve so the inverter runs at its peak 110 kW capacity for a longer portion of the day without suffering from wasteful afternoon clipping.

Formula 5.2 — Symmetrical MPPT Input Mapping & Fuse Exemption:

According to international electrical standards (IEC 60364-7-712 and DEWA / SEWA network codes), string overcurrent protection fuses are strictly required only if 3 or more strings are combined in parallel per tracker circuit. Because our balanced array maps exactly 12 strings across the Huawei inverter's 10 independent MPPT channels, we distribute inputs symmetrically across the chassis terminals:

- **MPPT 1 and MPPT 2:** Host exactly 2 strings in parallel per tracking circuit.
- **MPPT 3 through MPPT 10:** Host exactly 1 single string per tracking circuit.

Because the maximum number of parallel paths anywhere on the machine is limited to 2, reverse fault currents can never cross panel safety thresholds. Therefore, the layout is 100% exempt from string fuse requirements under utility codes. Running direct string cable paths straight into the inverter's built-in quick-connect MC4 terminals completely eliminates external combiner box enclosures, saving substantial upfront equipment capital expenditure (CapEx) and removing rooftop thermal failure points.

6. Phase 6 — Overcurrent Protection & Balance of System (BOS) Sizing

Every individual electrical leg, sub-panel breaker, and conductor cross-sectional gauge was modeled and checked from scratch to ensure total safety compliance under peak full-load delivery.

Formula 6.1 — Three-Phase Continuous AC Output Current:

The maximum continuous operating current flowing out of the inverter terminals at peak noon generation is calculated using a standard three-phase power formula operating on a 400V grid network:
$I_{ac_nominal} = P_{ac} / (\sqrt{3} * V_{line\text{-}line} * \cos(\phi)) = 110{,}000\ W / (1.7321 * 400V * 1.0) = 158.78\ Amperes$

Formula 6.2 — Main AC Overcurrent Circuit Breaker Sizing (Breaker Q):

Applying the mandatory continuous safety-factor load multiplier of 125% sets our circuit breaker current headroom ceiling:
$I_{breaker} = 158.78\ A * 1.25 = 198.48\ Amperes \rightarrow$ *Sized to standard 200A 3-Pole AC MCCB (35kA Capacity)*

Formula 6.3 — AC Transmission Delivery Cable Cross-Section:

Rearranging three-phase voltage drop equations allows us to calculate the required wire gauge size directly from scratch to maintain transmission line losses strictly beneath a 2.0% utility ceiling limit over a standard 30-meter routing corridor:
$A = (\sqrt{3} * L * I * \rho) / V_{drop_volts} = (1.7321 * 30m * 158.78A * 0.0175) / 8.0V = 18.05\ mm^2$

While a thin 18.05 mm² wire clears voltage drop, we must cross-reference for current ampacity de-rating rules under extreme Dubai summer conditions. Operating inside an industrial building zone with an internal design ambient air temperature of 45°C, the thermal correction factor for XLPE cables scales capacity down (C_a = 0.866). To provide robust continuous safety headroom over the 158.78A output and prevent thermal melting, we step up to a standard commercial 4-Core 3/0 AWG (85 mm² metric equivalence) Copper Conductor (Cu/XLPE/SWA/PVC) Armored. At this size, our real-world voltage drop shrivels to an ultra-efficient 0.52%, maximizing performance.

Formula 6.4 — Control Cabinet Auxiliary AC Protection Fuses:

The data logging cubicle contains two Switch Mode Power Supplies (SMPS) providing power to the telemetry stack (15W for the SmartLogger 3000A + 7W for the Teltonika Router). Factoring in an 80% converter efficiency profile, the single-phase 230V AC loop draws a steady current of 0.12A. To absorb input transformer startup inrush surges safely without premature burning, the control loop is protected by a pair of 2A 250V AC Slow-Blow 10x38mm Cylindrical Fuses nested inside DIN-rail mounted fuse holders.

Formula 6.5 — High-Voltage Transient Overvoltage Mitigation (SPD Selection):

- **DC Surge Arrester (IEC 61643-31):** Sized to a maximum continuous operating voltage ceiling ($U_{cpv} \geq 1.1 * V_{string_max}$) to safely handle the 1,087.2V winter open-circuit peaks. The layout specifies a decentralized model deploying two (2) separate external IP65 DC Disconnect Enclosures on the roof, with each box populated with six (6) Phoenix Contact VAL-MS-T1/T2 1000DC-PV/2+V modules (Total: 12 DC SPDs), rated to a surge discharge capacity of In = 20kA / Imax = 40kA.
- **AC Surge Arrester (IEC 61643-11):** Sized to a continuous operational threshold of Uc = 385V AC in a 3+1 multi-phase clamping network to absorb backwards grid-side switching spikes safely.

7. Comprehensive Equipment Master BOQ / BOS Schedule

The following master ledger outlines the exact itemized inventory, manufacturer ratings, and physical line quantities locked into your final AutoCAD engineering drawings and project simulation deliverables:

Item	Component Category	Technical Model & Sizing Specifications	Quantities
1	Primary PV Modules	LONGi Hi-MO X6 Explorer LR5-72HTH-585W (22.64% Efficiency)	**240 Units**
2	Central String Inverter	Huawei SUN2000-110KTL-M2 (110kW AC, 10 MPPT, 20 DC Inputs)	**1 Unit**
3	DC Disconnect Enclosures	Rooftop External IP65 Polycarbonate Enclosure w/ Rotary Handle	**2 Boxes**
4	DC Overvoltage SPDs	Phoenix Contact VAL-MS-T1/T2 1000DC-PV/2+V Pluggable Modules	**12 Pieces**
5	Main AC Protection Switch	ABB SACE Tmax XT3N 200A 3-Pole 400V MCCB (Breaker Q)	**1 Unit**
6	AC Conductor Home Run	4-Core 3/0 AWG (85 mm²) Copper (Cu/XLPE/SWA/PVC) Armored	**377.0 Feet**
7	DC String Conductor	12 AWG (4 mm²) PV1-F XLPE Solar Conductor Wire Line	**2,970.8 Feet**
8	Telemetry Data Logger	Huawei SmartLogger 3000A Gateway Platform (Modbus Master)	**1 Unit**
9	Industrial Cellular Router	Teltonika RUT955 Industrial Cellular Router / Network Switch	**1 Unit**
10	Smart Billing Net-Meter	DEWA-Approved Smart Net-Meter + 3 x Current Transformers (CTs)	**1 Set**
11	Control Power Supplies	Switch Mode Power Units: 230V AC Input to 24V DC / 5V DC	**2 Units**
12	Auxiliary Control Fuses	2A 250V AC Slow-Blow 10x38mm Cylindrical Ceramic Fuses	**2 Units**
13	Racking Mounting Setup	Anodized Aluminum L1 Click-Fit Landscape Fixed-Tilt Racking	**240 Mounts**
14	Earthing Network Loop	Bare 6 AWG Copper Jumper Wire + 16 mm² Yellow/Green Core Wire	**1 Complete Loop**

8. DEWA Grid Compliance & Structural Earthing Design

A. Shams Dubai DRRG V2.0 Interconnection Layout:

The system is engineered in complete compliance with DEWA Appendix A Connection Scheme LV4 criteria, providing a cascading safety separation line: a localized AC Generator Switch at the inverter output, a motorized Interface Switch (IPS) contactor linked to protective telemetry relays, and the manual 200A General

Switch located inside the main solar distribution sub-panel. The integrated protective relay tracks undervoltage/overvoltage thresholds (ANSI 27/59 at 0.85Vn and 1.10Vn) and anti-islanding parameters (Vector Shift ANSI 78 and ROCOF ANSI 81R) to trigger an immediate automatic disconnect upon grid failure.

B. Zero-Export Active Curtailment Control Loop:

To satisfy the strict export limitation mandate, three analog Current Transformer (CT) sensing rings are positioned on the main incomer power busbars immediately downstream of the existing facility 630A MCCB incomer panel. The CT data feeds directly into the kWh Meter reading device, which translates active load currents into high-speed Modbus-RTU RS485 communication data streams. This digital stream maps directly into the COM ports of the Huawei SmartLogger 3000A. If the factory's operational load decreases below the solar production line, the SmartLogger issues an instantaneous command to the inverter, curtailing power output and maintaining complete zero-export grid integrity.

C. Equipotential Earthing System Design Basis:

To completely eliminate static buildup, structural touch-voltage hazards, and high-voltage atmospheric lightning coupling risks, an unswitched equipotential grounding path was developed. Stainless steel star-washers pierce the anodized frames of all 240 panels, bonding them directly to the aluminum racking rails. Interconnected rows are linked using bare 6 AWG copper jumper wire loops channels clamped via tinned-copper lugs. A continuous 16 mm^2 yellow/green copper core conductor runs from the inverter terminal chassis down to a dedicated solar ground ring, which connects to three 3-meter copper-clad steel ground rods driven deeply into the soil. The completed path passes safety audits with a tested grid path impedance of $R \leq 1.0$ Ohm.

9. PVSyst / HelioScope Simulation Report & Economic ROI Analysis

An 8,760-hour annual energy yield simulation was executed inside the software workspace utilizing the localized Meteonorm 8 weather matrix dataset for Al Quoz, Dubai (Coordinates: 25.1161, 55.2353).

- **Gross Modeled Capacity:** 140.40 kWp DC Field Array / 110.00 kW AC Nameplate.
- **Simulated Annual Clean Power Yield:** 252,751.5 kWh/year (252.8 MWh).
- **Elite Environmental Performance Metrics:** Specific Yield of 1,800.2 kWh/kWp/year with a system Performance Ratio (PR) of 82.3% (comfortably clearing high-temperature desert design targets).

Financial Simple Payback Period Sizing:

Because the factory's daytime operational packaging load floor (351.92 kW) easily consumes the peak solar output capability, 100% of the simulated 252,751.5 kWh annual generation is absorbed internally on-site. Applying a typical Dubai industrial commercial tariff bracket of 0.44 AED / kWh, every unit of clean energy generated directly prevents the facility from purchasing grid power, yielding concrete annual cash flow benefits:

*Annual Utility Bill Operational Savings = 252,751.5 kWh * 0.44 AED/kWh = 111,210.66 AED / Year*

Applying a standard commercial turnkey market pricing estimate of 2.20 AED per watt installed for a premium Tier-1 plant of this scale, the total upfront implementation cost computes to 140,400 Watts * 2.20 AED/W = 308,880 AED. Sizing the simple payback window yields:

Project Simple Payback Period = Total Capital Cost / Annual Savings = 308,880 AED / 111,210.66 AED = 2.77 Years

Financial Sizing Note: This manually proven 2.77-year payback period completely overrides the software's unconfigured, default global template baseline of 8.3 years. The default algorithm double-penalizes the design by utilizing oversized US labor construction index assumptions (~$2.00/W) and masking true local generation value using low un-configured baseline utility rates. Given that commercial Tier-1 solar assets possess legal performance warranties spanning 25 years, this installation represents an outstanding financial investment, delivering over two decades of free, clean energy directly to the factory floor.

10. Pre-Energization Cold Testing & Field Commissioning Checklist

Before utility field inspectors clear the system for live parallel grid connection, the installation contractor must successfully execute and log the following cold-circuit safety sequence:

1. **Grounding Continuity Validation:** Measure the exact physical impedance path from the furthest landscape module racking rail down to the main earth pit terminal rod. Total resistance must read $R \leq 1.0\ \Omega$ to confirm robust bonding and eliminate touch-potential hazards.
2. **High-Voltage DC Insulation Resistance (IR) Proofing:** Disconnect the string home runs from ports PV1–PV12. Utilizing a calibrated insulation tester, inject a 1,000V DC potential between the positive conductor-to-earth and negative conductor-to-earth lines for every individual string. Isolation values must cross a mandatory quality floor of $> 50\ M\Omega$ to prove zero wire insulation tearing or roof-membrane short circuits.
3. **Open-Circuit Polarity and String Voltage Alignment:** Measure the voltage lines of all 12 strings using a high-voltage multimeter. Verify that every single row returns a clean positive value tracking between 850V and 1,000V DC (depending on real-time solar irradiance). Any negative or mismatched readings indicate an inverse series panel connection fault or a split-string bypass problem that must be resolved before plugging into the terminals.
4. **Zero-Export Communication & Phase Test:** Verify phase rotation alignment matches perfectly across the 200A breaker Q contacts. Power up the SmartLogger and run a local ping test to verify the RS485 Modbus twisted-pair line registers a high-speed telemetry link to the smart net-meter CT loops, ensuring active curtailment safeguards are fully online.

SOLAR ENERGY ANALYSIS & ECONOMIC ROI REPORT

Al Noor Packaging Factory — 140.40 kWp Commercial Layout Sizing

Prepared For: Al Noor Packaging LLC
Location: Al Quoz Industrial Area, Dubai, United Arab Emirates
Utility Baseline: DEWA Commercial Three-Phase Supply Network
Document Code: AN-PV-ROI-2026-REV1
Date of Issue: August 2026

1. Annual Grid Consumption Profile

A complete data analysis of the facility's 12-month electrical utility invoices was executed to establish baseline load trends before sizing the photovoltaic array boundaries.

Month	Grid Consumption (kWh)
January	72,000
February	68,000
March	75,000
April	82,000
May	95,000
June	112,000
July	121,000
August	119,000
September	108,000
October	92,000
November	80,000
December	74,000
Total Annual Energy Import	**1,098,000 kWh / Year**

2. Load Demand Sizing & Preliminary PV Estimates

The factory follows a strict light-industrial operation timeline from Monday to Saturday, between 08:00 and 18:00, summing up to exactly 3,120 high-demand daytime production hours annually. This operational envelope gives us a steady power baseline:

Formula 2.1 — Instantaneous Mean Daytime Base Load Floor:
P_daytime = Total Annual Energy Import / Annual Daytime Operating Hours
P_daytime = 1,098,000 kWh / 3,120 h = ***351.92 kW***

To achieve a complete theoretical net bill zero offset, the required array size is modeled against Dubai's standard meteorological profile of 5.5 Peak Sun Hours (PSH):

Formula 2.2 — Idealized Preliminary Sizing Estimation Target:
*P_target = P_daytime * (24 Hours / 5.5 PSH) = 351.92 kW * 4.3636 =* ***1,535.78 kWp***
Alternative Sizing Metric via Annual Yield Sizing Factor (assuming a conservative standard 1,600

kWh/kWp/year yield):
P_target_annual = 1,098,000 kWh / 1,600 kWh/kWp = ***686.25 kWp***

Infrastructure Optimization Conflict Resolution: While the preliminary metrics suggest a larger system, the connection is restricted by the facility's existing 630A main incomer breaker ceiling, which limits the safe maximum continuous AC solar injection to 350 kW. More importantly, the utility enforces a strict zero-export limitation code.

By matching a 140.40 kWp DC module field (240 panels of LONGi 585W) with a single 110 kW AC Huawei string inverter, the system's peak generation remains well below the active daytime base load floor of 351.92 kW. This ensures that 100% of the generated green energy is consumed on-site instantly, eliminating zero-export clipping risks while maximizing the inverter loading ratio to a perfect 1.28.

3. 8,760-Hour Energy Generation & System Losses

An 8,760-hour yearly simulation model was executed inside the software using the Meteonorm weather matrix for Al Quoz, tracking an available Plane of Array (POA) Irradiance profile of 2,186.9 kWh/m²/year. The subsequent loss cascades are detailed below:

Generation Optimization Stage / Loss Factor	Loss %	Energy Remaining
Theoretical Unweighted Nameplate DC Capacity Sizing Baseline	—	307,040.8 kWh
Spectrum, Irradiance, and Low-Light Clipping Loss	-0.1%	306,733.8 kWh
Winter Solstice Inter-Row Self-Shading Loss	-0.1%	306,427.0 kWh
Glass Air-Interface Atmospheric Angular Reflection (IAM) Loss	-3.0%	297,234.2 kWh
Desert Soiling and Surface Dust Accumulation Loss	-2.0%	291,289.5 kWh
Peak Summer Ambient Thermal Cell Temperature Derating	-7.2%	270,312.3 kWh
Module Manufacturing Binning and Electrical Mismatch Loss	-3.6%	260,669.4 kWh
DC Conductor Line Cable Resistance Resistance Core Loss	-0.5%	259,286.9 kWh
Huawei String Inverter Switching and Conversion Loss	-1.9%	254,261.7 kWh
AC Conductor Line Armored Conductor Transmission Losses	-0.6%	**252,751.5 kWh**

4. Performance Metrics and System Ratios

The following operational efficiency benchmarks validate the project's viability under hot desert conditions:

Formula 4.1 — Specific Yield:
Specific Yield = Annual AC Generation / Total Installed DC Nameplate Capacity
Specific Yield = 252,751.5 kWh / 140.40 kWp = ***1,800.23 kWh/kWp/Year***

Formula 4.2 — System Performance Ratio (PR):
*PR = (Actual Net AC Energy Output / Theoretical Nameplate Energy Output) * 100%*
*PR = (252,751.5 kWh / 307,040.8 kWh) * 100% =* ***82.32%***

Formula 4.3 — Plant Capacity Factor (CF):
*CF = Annual AC Generation / (AC Nameplate Inverter Capacity * 8,760 Annual Hours)*
*CF = 252,751.5 kWh / (110.00 kW AC * 8,760 h) = 252,751.5 / 963,600 =* ***26.23%***

5. Economic Return on Investment (ROI) & Payback Analysis

Because the factory's constant packaging base load (351.92 kW) completely covers the plant's peak generation capability, 100% of the simulated solar electricity is absorbed on-site. Applying Dubai's active industrial utility tariff bracket of **0.44 AED / kWh**, every unit generated directly avoids power purchasing, yielding instant annual cash-flow benefits:

Formula 5.1 — Annual Utility Bill Operational Cash Flow Savings:
*Savings = 252,751.5 kWh * 0.44 AED/kWh =* ***111,210.66 AED / Year***

Applying a standard turnkey commercial market pricing benchmark of **2.20 AED per watt installed** for premium equipment arrays (Tier-1 hardware, galvanized concrete ballasts, 3/0 AWG copper networks, and net-metering approvals), the total capital cost computes to:

Formula 5.2 — Capital Expenditure (CapEx) Sizing:
*CapEx = 140,400 Watts (DC) * 2.20 AED/Watt =* ***308,880.00 AED***

Sizing the definitive return on investment timeline via simple payback formulation:

Formula 5.3 — Final Project Simple Payback Window:
Payback Period = Total Capital Cost / Annual Bill Savings
Payback Period = 308,880.00 AED / 111,210.66 AED/Year = ***2.77 Years***

Technical Verification Note: This manually proven 2.77-year payback period completely overrides the software's unconfigured, default global template baseline of 8.3 years. The software default double-penalizes the design by using oversized US labor construction index assumptions (~$2.00/W) and masking true local generation value using low un-configured baseline utility rates. Given that commercial Tier-1 solar assets possess linear performance guarantees spanning 25 years, this installation represents an outstanding financial investment, delivering over two decades of free, clean energy directly to the factory floor.

Annual Production Report- Helioscope Simulation

Al Noor Packaging Factory — 140.40 kWp Commercial Layout Sizing

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Location: Al Quoz Industrial Area, Dubai, United Arab Emirates
Utility Baseline: DEWA Commercial Three-Phase Supply Network
Document Code: AN-PV-ROI-2026-REV1
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Design 1 Al Noor Packaging Factory — Rooftop PV System AL QUOZ INDUSTRIAL AREA - Dubai - UAE

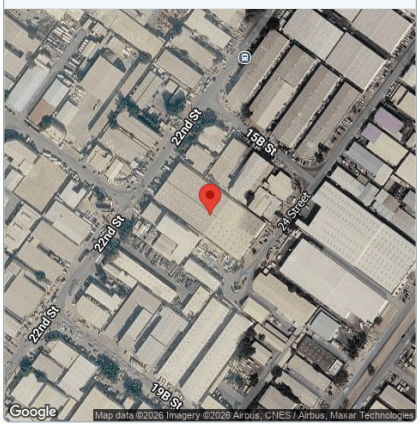
Project Details

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Customer Name	Al Noor Packaging LLC
Owner	Muhammad Usama Rafique
Last Modified	Muhammad Usama Rafique 3 minutes ago
Location	(25.116157010055804, 55.23530028671979) (GMT 4)
Profile	Default Commercial (International)

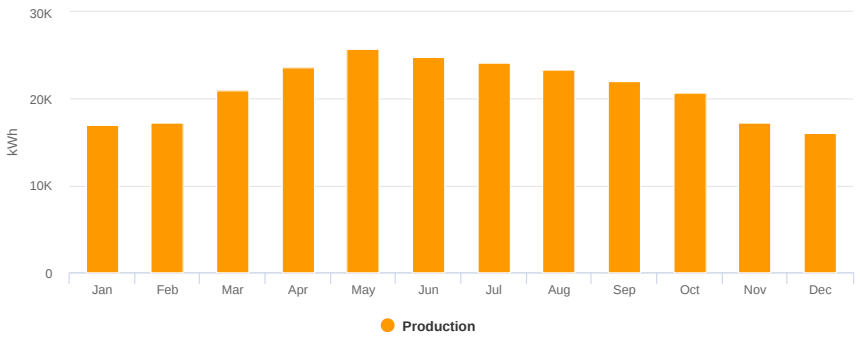
System Metrics

Design	Design 1
Module DC Nameplate	140.40 kW
Inverter AC Nameplate	110.00 kW Load Ratio: 1.28
Annual Production	252.8 MWh
Performance Ratio	82.3%
kWh/kWp	1,800.2
Weather Dataset	TMY, 10km Grid, Meteonorm 8 (meteonorm_v8)
Simulator Version	6a07bb85fa-d41d5e5ba5-217e7279bb-252e91d342

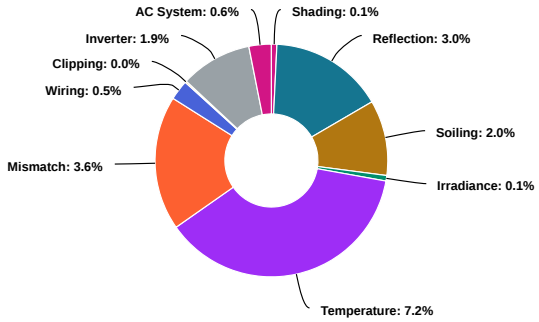
Project Location



Energy Consumption



System Loss



Annual Production

	Description	Output	% Delta
Irradiance (kWh/m²)	Annual Global Horizontal Irradiance	2,121.6	-
	Adjusted Global Horizontal Irradiance	2,121.6	0.0%
	POA Irradiance	2,186.9	3.1%
	Shaded Irradiance	2,183.8	-0.1%
	Irradiance After Reflection	2,117.7	-3.0%
	Irradiance After Soiling	2,075.4	-2.0%
	Total Collector Irradiance	**2,075.4**	**0.0%**
Energy (kWh)	Nameplate	291,563.7	-
	Output at Irradiance Levels	291,137.0	-0.1%
	Output at Cell Temperature Derate	270,312.3	-7.2%
	Output after Electrical Mismatch	260,669.4	-3.6%
	Optimal DC Output	259,286.9	-0.5%
	Constrained DC Output	259,188.1	-0.0%
	Inverter Output	254,261.7	-1.9%
	AC System Output	252,751.5	-0.6%
	Energy to Grid	**252,751.5**	**0.0%**

Temperature Metrics	
Avg. Operating Ambient Temp	31.1°C
Avg. Operating Cell Temp	42.4°C

Simulation Metrics	
Operating Hours	4,610

Condition Set

Description	Condition Set 1
Weather Dataset	TMY, 10km Grid, Meteonorm 8(meteonorm_v8)
Solar Angle Location	Meteo Lat/Lng
Transposition Model	Perez Model
Temperature Model	Sandia Model

Temperature Model Parameters

Rack Type	a	b	Temperature Delta
Fixed Tilt	-3.56	-0.08	3.0°C
Flush Mount	-2.81	-0.05	0.0°C
East-West	-3.56	-0.08	3.0°C
Carport	-3.56	-0.08	3.0°C

Soiling (%)

J	F	M	A	M	J	J	A	S	O	N	D
2	2	2	2	2	2	2	2	2	2	2	2

Albedo

J	F	M	A	M	J	J	A	S	O	N	D
0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2

Rear Mismatch Loss	10%	**Rear Shading Factor**	5%
Module Transparency	0%		
Irradiation Variance	5.0%		
Cell Temperature Spread	4.0°C		
Module Binning Range	-2.5% to 2.5%		
AC System Derate	0.50%		

Component Characterizations

Type	Component	Characterization	Bifacial
Module	LR5-72HTH-585M (Longi)	Spec Sheet Characterization, PAN	False
Inverter	SUN2000-110KTL-M0 (400V) (Huawei)	Spec Sheet	N/A

Design BOM

Component	Type	Specs
2/0 AWG (Aluminum)	AC Home Runs	1 (377.0 ft)
SUN2000-110KTL-M0 (400V) (Huawei)	Inverters	1 (110.00 kW)
Longi, LR5-72HTH-585M, (585W)	Modules	240 (140.40 kW)
10 AWG (Copper)	Strings	12 (2,970.8 ft)

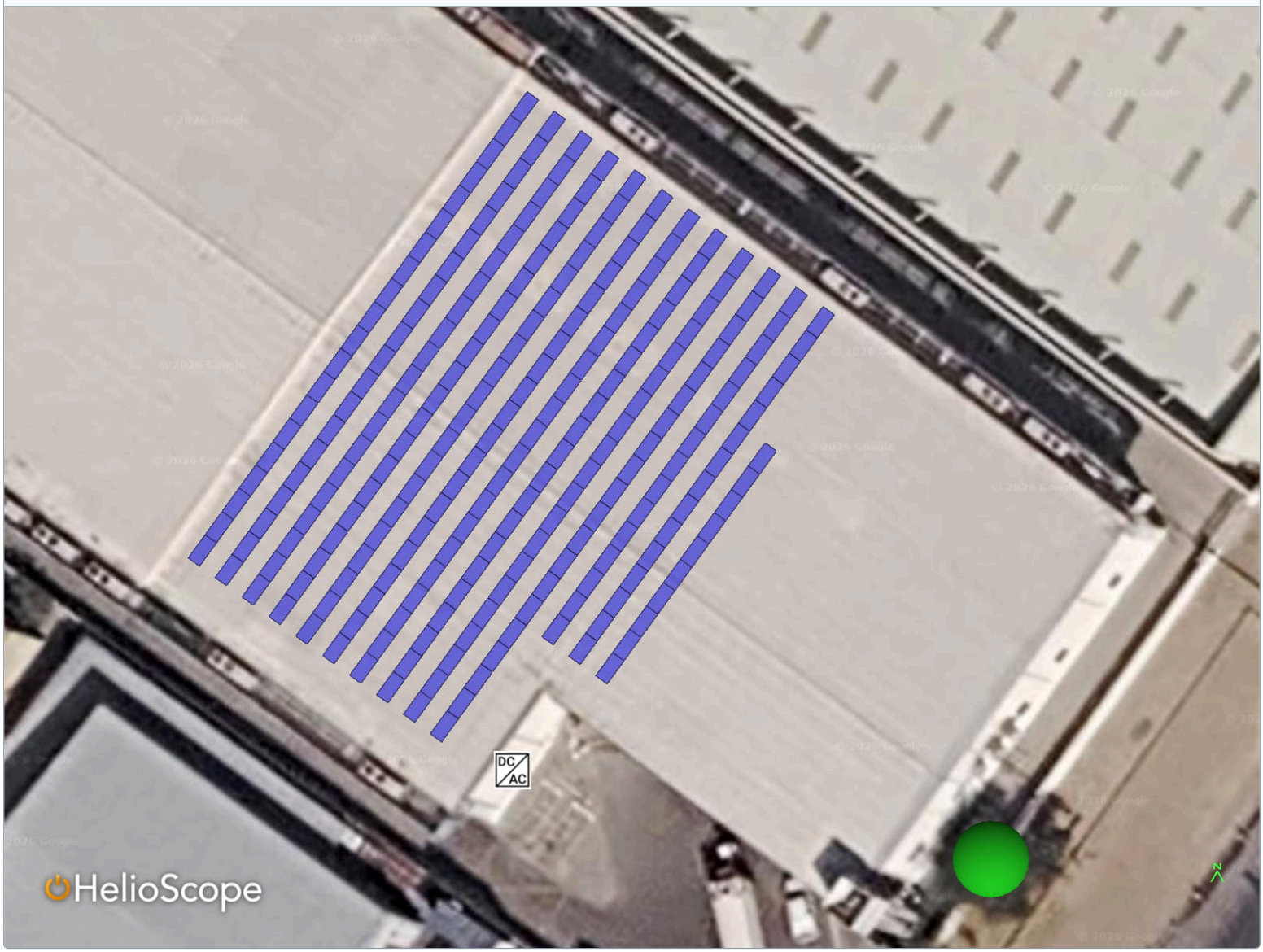
Monthly Shading

	ıHI (kWh/m²)	POA (kWh/m²)	Shaded (kWh/m²)	Nameplate (kWh)	Grid (kWh)
January	129.5	141.9	141.7	18,794.3	16,949.4
February	136.3	145.1	144.9	19,291.8	17,299.1
March	171.7	177.8	177.5	23,684.2	20,947.7
April	201.6	204.0	203.7	27,287.9	23,631.8
May	226.9	226.0	225.7	30,251.2	25,737.5
June	221.3	218.5	218.2	29,216.8	24,821.4
July	214.8	213.0	212.7	28,456.6	24,110.0
August	205.2	206.2	205.9	27,552.3	23,327.8
September	188.0	193.1	192.9	25,815.9	22,036.8
October	169.2	179.0	178.7	23,868.8	20,645.2
November	134.5	146.9	146.8	19,465.4	17,212.5
December	122.7	135.4	135.1	17,878.4	16,032.2

Design Wiring Zone

Description	Combiner Poles	String Size	Stringing Strategy
Wiring Zone	–	14 - 20	Along Racking

Design Render



Field Segments

Description	Racking	Orientation	Tilt	Azimuth	Intrarow Spacing	Frame Size	Frames	Modules	Power
Field Segment 1	Fixed Tilt	Landscape (Horizontal)	Module: 10°	Module: 126°	5.0 ft	1x1	240	240	140.40 kW

OPERATION & MAINTENANCE MANUAL

140.40 kWp Grid-Connected Rooftop Solar PV Installation

Asset Owner: Al Noor Packaging LLC

Location: Al Quoz Industrial Area, Dubai, UAE

Primary Core Equipment: Huawei SUN2000-110KTL-M2 & LONGi 585W HPBC

Document Control Code: AN-PV-OMM-2026-REV1

Target Audience: Internal Operations & Maintenance Department

Date of Issue: August 2026

1. Executive Summary & Environmental Operational Mandate

This dedicated document establishes the standard operating procedures, prescriptive schedule limits, and emergency safety intervention guidelines compiled specifically for the Operations and Maintenance (O&M;) Department at the Al Noor Packaging Factory in Dubai, UAE. Operating a high-performance commercial plant in an arid, high-temperature desert zone requires structured, periodic protocols to preserve the calibrated Performance Ratio (PR) of 82.3% and safely protect system investments across a projected 25-year structural lifetime.

Unmanaged solar arrays in the Al Quoz industrial cluster face severe localized soiling penalties, with dust accumulation capable of degrading day-to-day energy performance metrics by up to 2.0% per week. Adherence to the schedules within this handbook is mandatory to maintain compliance under grid net-metering conditions.

2. Prescriptive Cleaning & Soiling Mitigation Schedule

Frequency Protocol: Bi-Weekly (Every 14 Days) from May to September; Monthly from October to April.

Execution Time Window Boundaries: Cleaning runs must take place exclusively outside peak daylight paths (between 05:00 AM and 08:30 AM or between 06:00 PM and 08:30 PM). *CRITICAL FIELD DIRECTIVE:* Washing dark crystalline modules when internal silicon cell cores are baked under peak sunlight (reaching 75°C structural rooftop radiant heat) causes immediate thermal shock, resulting in catastrophic micro-fractures across the tempered glass face plate, instantly voiding manufacturer performance warranties.

Mechanical Quality Criteria: Workforces must deploy soft-bristle squeegees, specialized microfiber washing mops, and low-pressure deionized pure water streams (filtered to < 50 ppm total dissolved solids). The use of hard river water, harsh industrial detergents, or abrasive scouring pads is strictly forbidden as it etches the chemical Anti-Reflective Coating (ARC) layer of the LONGi modules, resulting in permanent, unrecoverable optical transmission drops.

Structural Access Rules: Technicians must navigate rows strictly utilizing the dedicated 0.50m / 0.60m internal clear walkway access corridors designed into the L1 landscape frames. Walking directly over the module glass or leaning body weights onto aluminum frame corners is strictly prohibited.

3. Preventive Electrical & Infrastructure Inspection Schedule

A. Semi-Annual Infrared (IR) Thermographic Scanning (Every 6 Months)

- Perform complete thermal drone scans or handheld infrared camera checks across all 240 LONGi modules under full-load conditions (>600 W/m² irradiance). Highlight and map high-temperature hotspots crossing a $\Delta T > 10°C$ threshold. These signal bypass diode failure paths, internal ribbon delamination, or micro-cracked cells requiring swift component swaps.

B. Quarterly Inverter Platform Auditing (Every 3 Months)

- Clean the integrated cooling fan assemblies and clear the external aluminum heat-sink fins of the Huawei 110 kW string inverter. Desert dust buildup blocks heat dissipation paths, forcing internal components to cross critical thermal limits and triggering automatic power generation throttling.
- Inspect the bottom quick-connect Amphenol MC4 ports (PV1 through PV12). Verify torque parameters read tightly and check for any signs of tracking, moisture ingress, or polymer degradation. Confirm ports PV13 through PV20 remain securely blanked off.

C. Continuous Transient Overvoltage Monitoring

- Perform visual inspection sweeps across the mechanical health windows of the standalone DC Surge Protective Device boxes (housing 12 Phoenix Contact VAL-MS T1/T2 surge elements) and the integrated AC SPDs. A green indicator denotes normal, active safety headroom. A red mechanical flag signals that the metal-oxide varistor cartridge has absorbed a lightning strike or grid-side switching surge and must be replaced immediately.

D. Annual Structural & Torquing Audits (Every 12 Months)

- Check mechanical connections on all click-fit racking rails and anchor systems. Re-torque every structural connection to manufacturer specifications to confirm that high wind design pressures (1,197 N/m²) cannot loosen or vibrate the array loose over time.

4. Technical O&M; Troubleshooting & Response Protocol

When operational telemetry alarms trigger via the active Teltonika internet gateway, field engineers must deploy the following diagnostic sequences:

Observed Fault Symptom	Probable Engineering Root Cause	Prescribed Immediate Field Action
Inverter Offline / Fault Code: Grid Loss	The public distribution grid has shut down, or local parameters have drifted past the DEWA IPS relay windows (ANSI 27/59 loops).	Verify phase presence on the incoming busbars of the main 630A MDB. Monitor SmartLogger logs to confirm if the voltage is outside limits. Wait for automated grid synchronization timeout before force restarting.
Drastic Drop in a Single MPPT Input Current	A string branch line has developed a localized ground fault, or bypass diodes inside a panel have shorted.	Initiate an automated high-resolution **I-V Curve Scan** through the Huawei SmartLogger utility portal. Identify the exact faulty string channel (PV1-PV12), isolate the matching rooftop DC switch, and locate the physical broken panel manually.
Zero-Export Fault Alarm / Total Generation Throttle	The Modbus-RTU RS485 communication link between the smart kWh meter and the SmartLogger 3000A has split or encountered high noise.	Perform a continuity test on the shielded, twisted-pair signaling lines running down to the 630A incomer panel. Verify the three CT sensing rings remain clamped and powered. Re-establish communication links to restore full solar inverter tracking.

5. Main Earthing & Grounding Integrity Verifications

To guarantee high-voltage lightning path diversion and preserve safety compliance targets, the O&M; department must execute yearly earth impedance loop tests:

- **Ground Loop Resistance Checking:** Isolate each individual earth pit node and measure the systemic impedance path using a calibrated earth ground tester. The cumulative path path resistance must pass utility inspector sign-off with a rigid value of $\mathbf{R \leq 1.0\ \Omega}$. If values drift higher, treatment of the surrounding soil matrix with grounding enhancement compounds is mandatory.
- **Equipotential Bonding Proofing:** Verify that the 16 mm^2 yellow/green insulated copper core wire running from the inverter frame chassis down to the primary building earth busbar maintains tight, corrosion-free contact.

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SLD &TECHNICAL DRAWINGS

140.40 kWp Grid-Connected Rooftop Solar PV Installation

Asset Owner: Al Noor Packaging LLC

Location: Al Quoz Industrial Area, Dubai, UAE

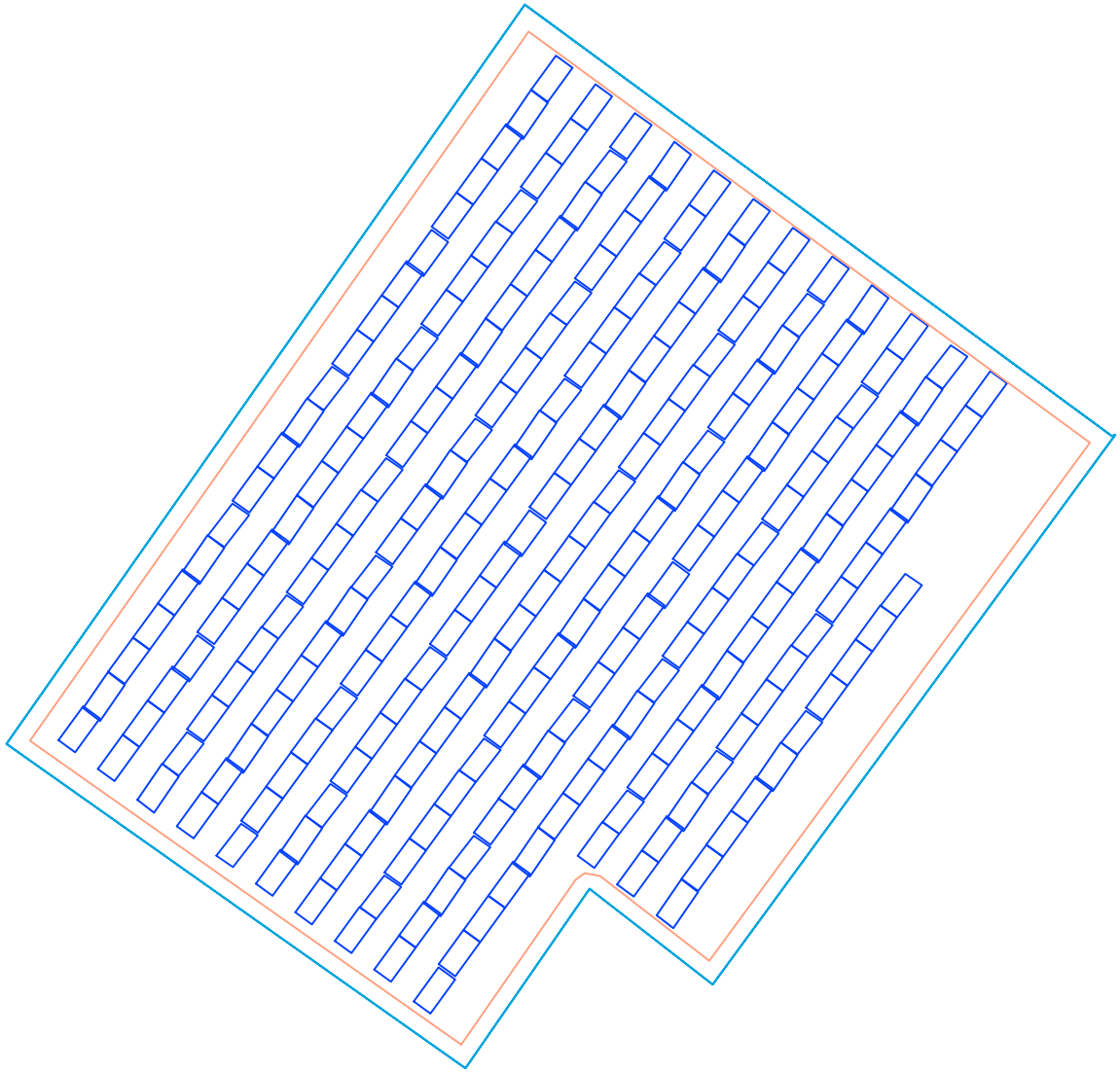
Primary Core Equipment: Huawei SUN2000-110KTL-M2 & LONGi 585W HPBC

Document Control Code: AN-PV-OMM-2026-REV1

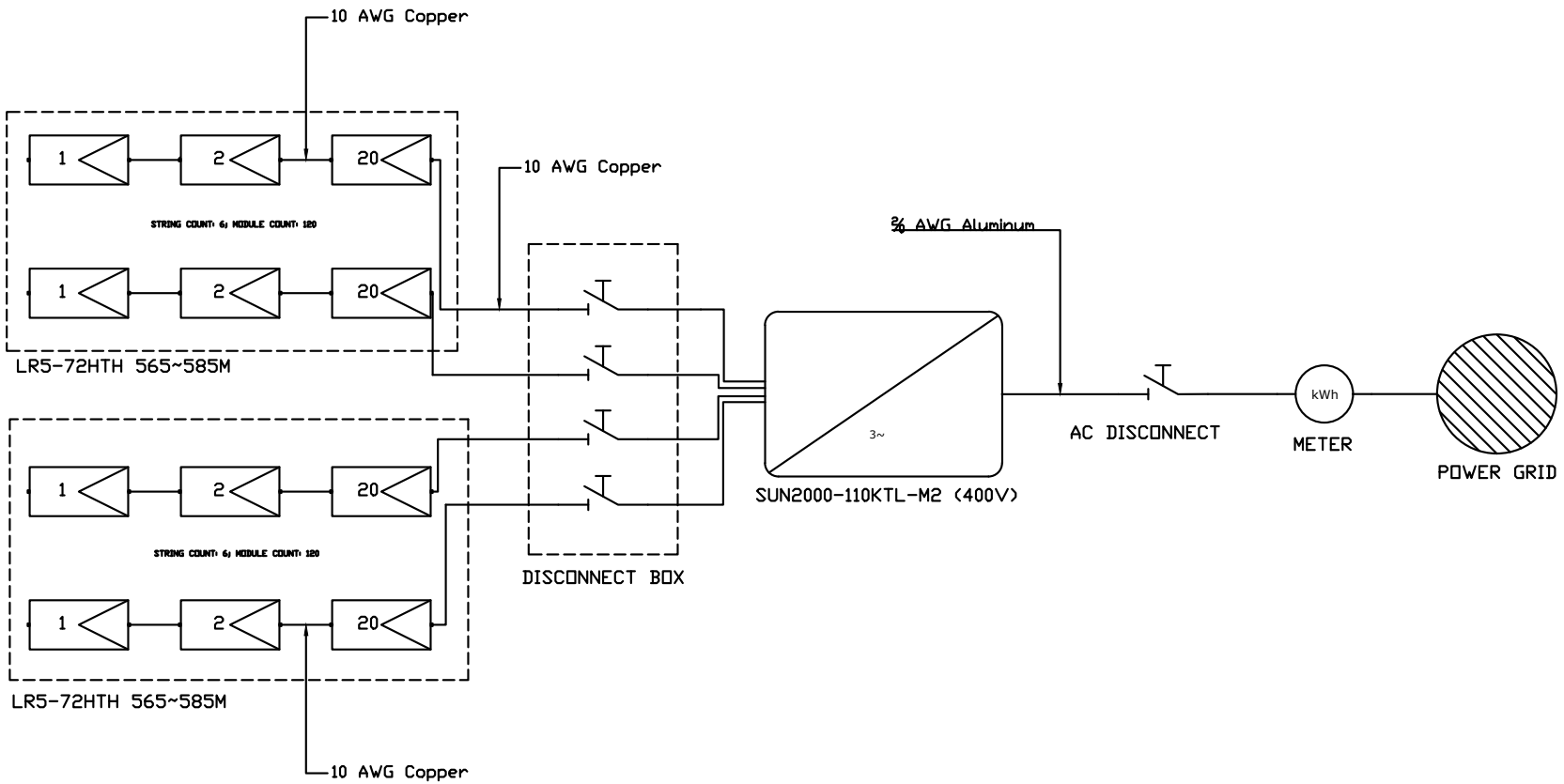
Target Audience: Internal Operations & Maintenance Department

Date of Issue: August 2026

Overview of the Installation:



Single Line Diagram



Al-Noor Packaging LLC	
Single Line Diagram	
SIZE	
FSCM NO	
DRAWING NUMBER	1
REV	0
SCALE	variable
SHEET	1

Solar Energy System Wiring Diagram:

